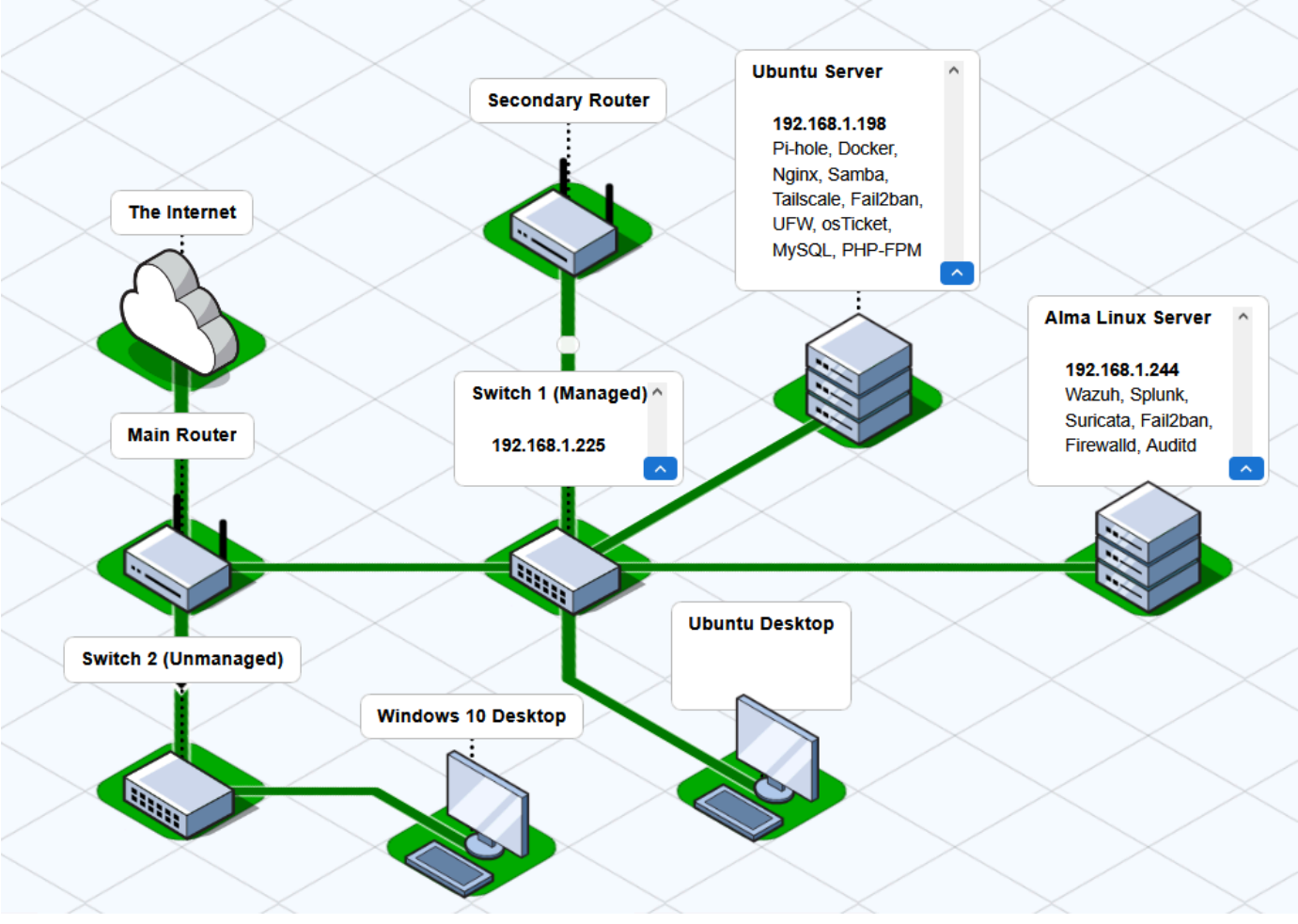
Eric Guzman 3/6/2026

IT Help Desk Ticketing System Lab: osTicket Deployment, Workflow Design, and Operations

Executive Summary: This report documents the deployment and configuration of a production-realistic IT help desk environment built on Ubuntu Server 25.04, using osTicket 1.18.3 as the central ticketing platform. The environment was provisioned on bare-metal hardware running an nginx web server, MySQL 8.4, and PHP 8.4 with no virtualization layer. The deployment covers the full operational lifecycle of a corporate help desk: ticket intake through a web portal, automatic routing by department and SLA tier, agent assignment, internal escalation between Tier 1 and Tier 2 support, and resolution documentation. Eight tickets were submitted across four departments spanning workstation issues, network troubleshooting, hardware provisioning, access management, and a simulated SEV-1 critical incident. Standard operating procedures were developed for each major workflow category. This report documents the environment architecture, configuration decisions, ticket workflows, and operational procedures end to end.

Topology:



Tools and Stack:

Tool	Version	Role
osTicket	v1.18.3	Help desk ticketing platform: ticket intake, routing, SLA enforcement, agent workflow, escalation
Ubuntu Server	25.04	Base OS: host operating system for all services
nginx	1.26.3	Web server: reverse proxy serving osTicket on port 8080
MySQL	8.4.7	Database backend: ticket data, user accounts, configuration
PHP-FPM	8.4	Application runtime: PHP processing via FastCGI for nginx

Skills Demonstrated:

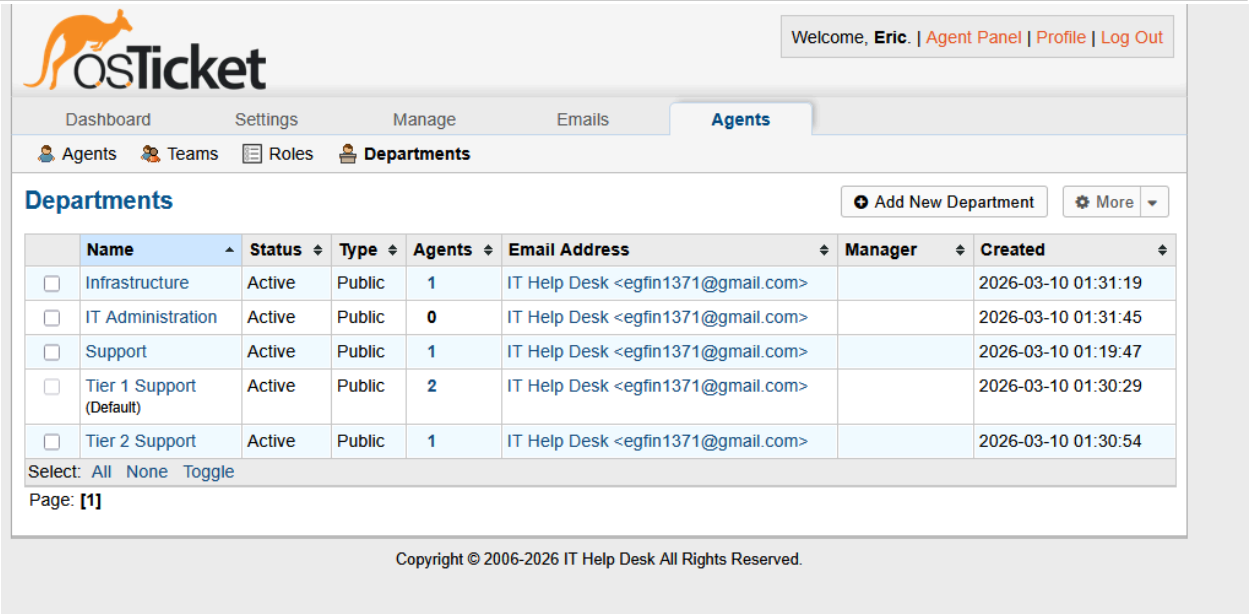
Skill	Detail
Linux administration	Service configuration, LVM expansion, systemctl, nginx vhost management, PHP-FPM tuning on Ubuntu Server 25.04
Help desk platform administration	osTicket department structure, SLA policy configuration, agent and team provisioning, help topic routing
Ticket workflow design	Tier 1 resolution, Tier 1 to Tier 2 escalation paths, SEV-1 critical incident handling, hardware provisioning workflow, access request management
SOP documentation	Written standard operating procedures for each major ticket category, internal note discipline, escalation handoff documentation
Troubleshooting	PHP-FPM socket authentication, MySQL caching_sha2_password compatibility, nginx FastCGI timeout configuration, LVM filesystem expansion

Section 1: Help Desk Structure and Configuration

The help desk environment was configured to reflect the organizational structure of a mid-size corporate IT support team. Four departments were created to handle different categories of work, with agents assigned to each department and SLA plans governing response windows by ticket severity. All configuration was performed through the osTicket admin panel after deployment.

1a. Department Structure

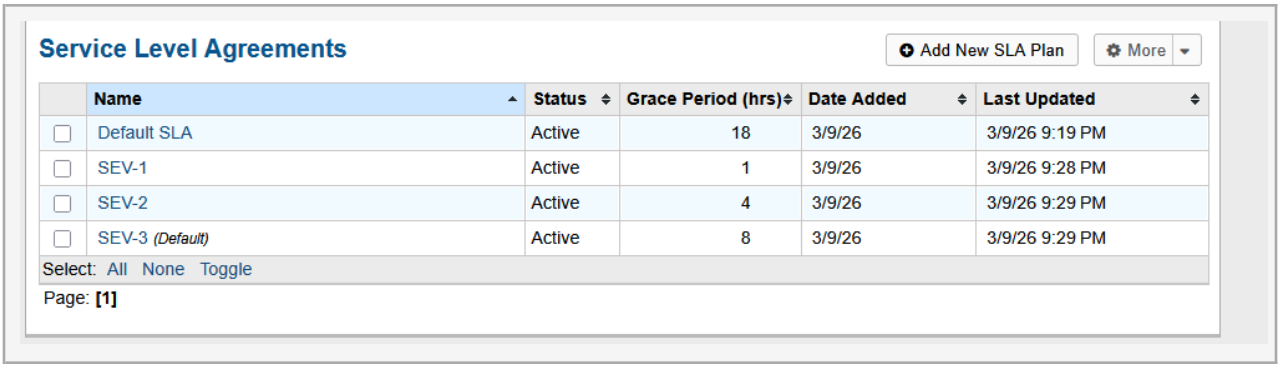
Department	SLA	Purpose
Tier 1 Support	SEV-3	First contact for all incoming tickets. Handles password resets, workstation issues, software requests, and conference room support.
Tier 2 Support	SEV-2	Escalated technical issues requiring deeper investigation. Handles network troubleshooting and critical incidents.
Infrastructure	SEV-2	Hardware provisioning, device imaging, and new hire equipment setup.
IT Administration	SEV-3	Access and permissions requests, account management, approval-gated workflows.



1b. SLA Plans

Three SLA tiers were configured to enforce response windows based on ticket severity. SLA plans are assigned at the help topic level so tickets are automatically tiered at submission without requiring manual triage for standard requests.

SLA Plan	Grace Period	Schedule	Intended Use
SEV-1	1 hour	24/7	Critical outages, system-wide failures, business-blocking incidents
SEV-2	4 hours	24/7	Hardware failures, network issues, escalated tickets requiring Tier 2 investigation
SEV-3	8 hours	Business Hours	Standard requests, software installs, access requests, conference room support



Service Level Agreements

Add New SLA Plan | More

Name	Status	Grace Period (hrs)	Date Added	Last Updated
Default SLA	Active	18	3/9/26	3/9/26 9:19 PM
SEV-1	Active	1	3/9/26	3/9/26 9:28 PM
SEV-2	Active	4	3/9/26	3/9/26 9:29 PM
SEV-3 *(Default)*	Active	8	3/9/26	3/9/26 9:29 PM

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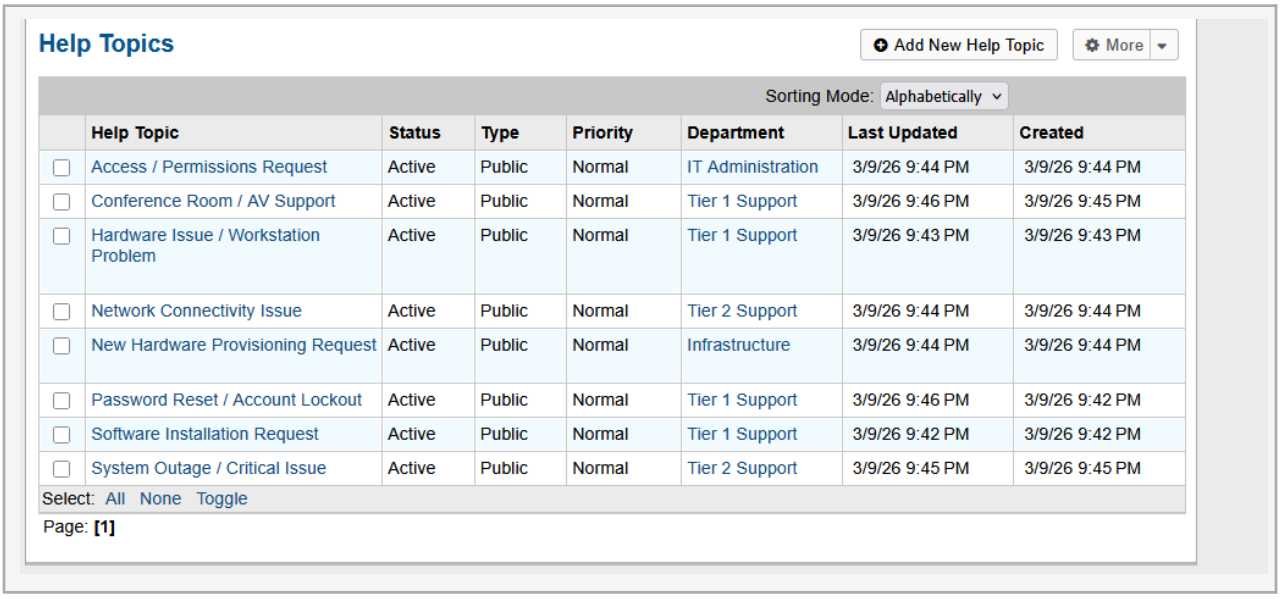
1c. Help Topics and Routing

Eight help topics were configured to cover the primary ticket categories in a corporate help desk environment. Each help topic maps to a specific department and SLA plan so tickets are automatically routed and tiered at submission.

Help Topic	Department	SLA
Password Reset / Account Lockout	Tier 1 Support	SEV-3

Help Topic	Department	SLA
Hardware Issue / Workstation Problem	Tier 1 Support	SEV-2
Software Installation Request	Tier 1 Support	SEV-3
Conference Room / AV Support	Tier 1 Support	SEV-3
Network Connectivity Issue	Tier 2 Support	SEV-2
System Outage / Critical Issue	Tier 2 Support	SEV-1
New Hardware Provisioning Request	Infrastructure	SEV-3
Access / Permissions Request	IT Administration	SEV-3



Help Topics

Add New Help Topic | More

Sorting Mode: Alphabetically

Help Topic	Status	Type	Priority	Department	Last Updated	Created
Access / Permissions Request	Active	Public	Normal	IT Administration	3/9/26 9:44 PM	3/9/26 9:44 PM
Conference Room / AV Support	Active	Public	Normal	Tier 1 Support	3/9/26 9:46 PM	3/9/26 9:45 PM
Hardware Issue / Workstation Problem	Active	Public	Normal	Tier 1 Support	3/9/26 9:43 PM	3/9/26 9:43 PM
Network Connectivity Issue	Active	Public	Normal	Tier 2 Support	3/9/26 9:44 PM	3/9/26 9:44 PM
New Hardware Provisioning Request	Active	Public	Normal	Infrastructure	3/9/26 9:44 PM	3/9/26 9:44 PM
Password Reset / Account Lockout	Active	Public	Normal	Tier 1 Support	3/9/26 9:46 PM	3/9/26 9:42 PM
Software Installation Request	Active	Public	Normal	Tier 1 Support	3/9/26 9:42 PM	3/9/26 9:42 PM
System Outage / Critical Issue	Active	Public	Normal	Tier 2 Support	3/9/26 9:45 PM	3/9/26 9:45 PM

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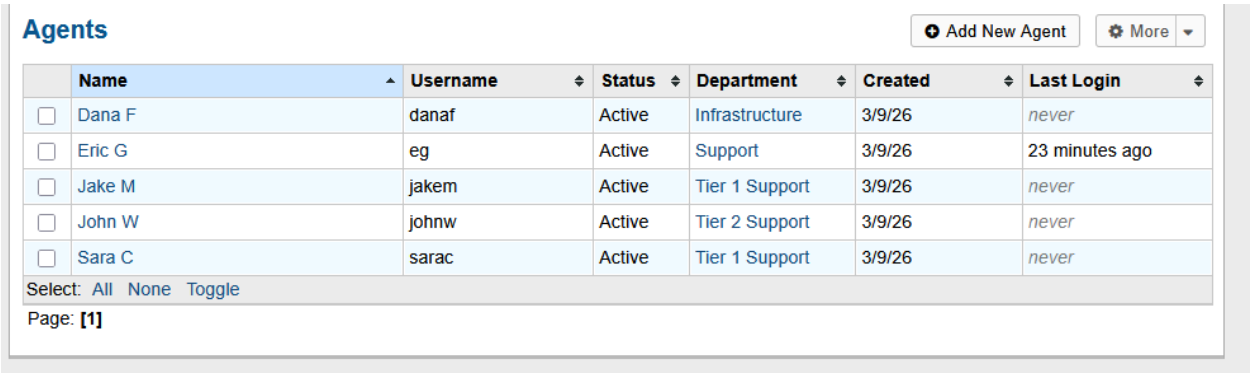
1d. Agent Roster

Four agents were provisioned across the support departments. Two Tier 1 technicians handle first-contact tickets and two senior agents are assigned to Tier 2 and Infrastructure for escalated work. Agents were also assigned to teams to support the escalation routing configuration.

Agent	Department	Team
Jake Morris	Tier 1 Support	Level 1 Response
Sara Chen	Tier 1 Support	Level 1 Response
John Webb	Tier 2 Support	Level 2 Escalation
Dana Flores	Infrastructure	Level 2 Escalation



Agents

Add New Agent | More

Name	Username	Status	Department	Created	Last Login
Dana F	danaf	Active	Infrastructure	3/9/26	never
Eric G	eg	Active	Support	3/9/26	23 minutes ago
Jake M	jakem	Active	Tier 1 Support	3/9/26	never
John W	johnw	Active	Tier 2 Support	3/9/26	never
Sara C	sarac	Active	Tier 1 Support	3/9/26	never

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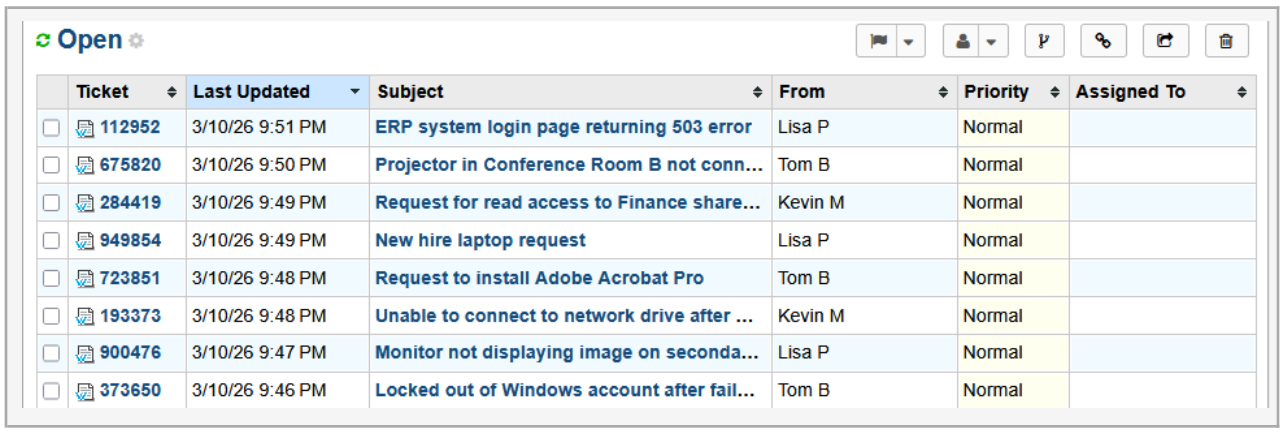
Section 2: Ticket Workflows

Eight tickets were submitted across all four departments. The following section documents each major workflow category, the routing behavior, and the actions taken within each ticket. Screenshots show representative workflows demonstrating Tier 1 resolution, Tier 1 to Tier 2 escalation, and SEV-1 critical incident handling.



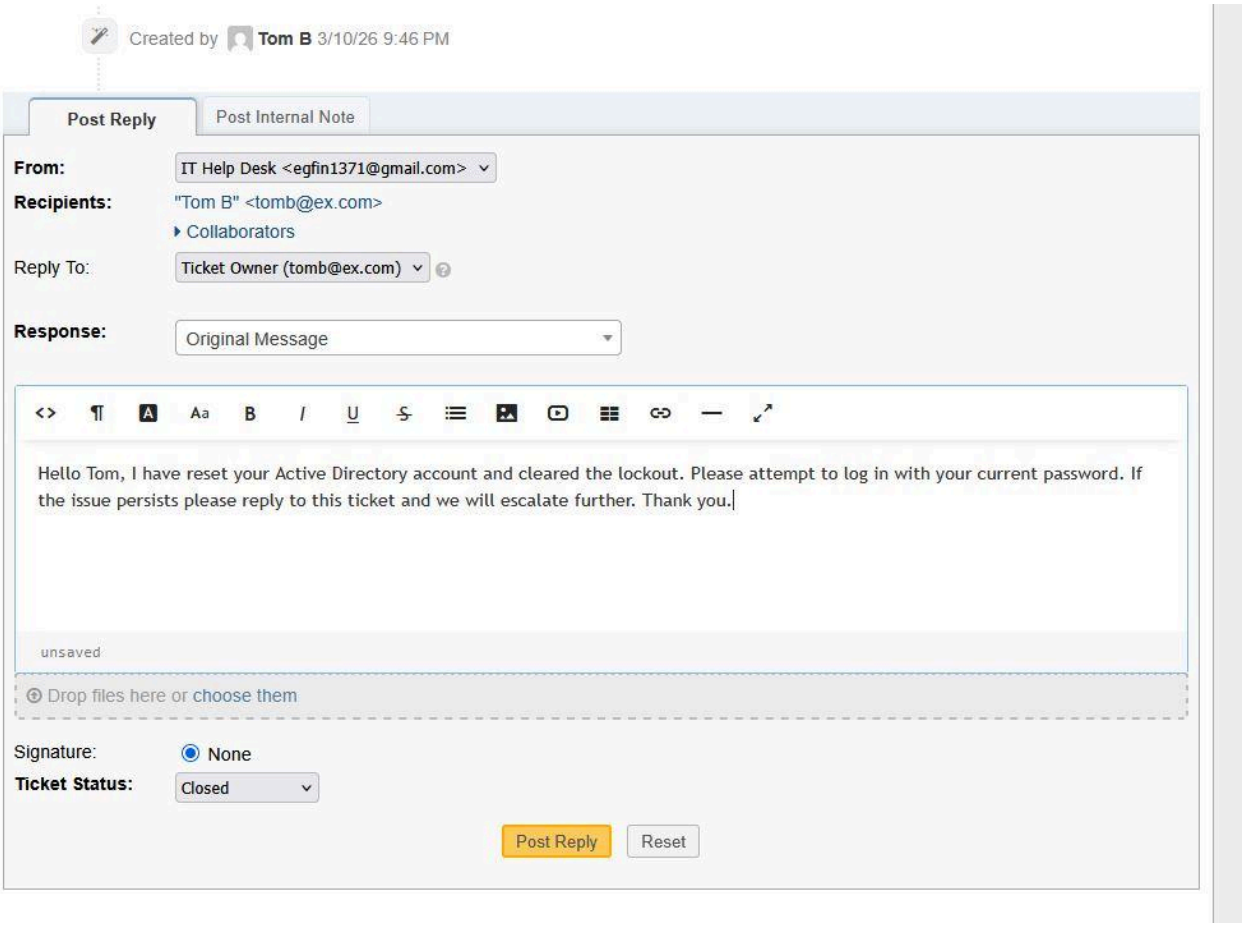
Open

Ticket	Last Updated	Subject	From	Priority	Assigned To
112952	3/10/26 9:51 PM	ERP system login page returning 503 error	Lisa P	Normal	
675820	3/10/26 9:50 PM	Projector in Conference Room B not conn...	Tom B	Normal	
284419	3/10/26 9:49 PM	Request for read access to Finance share...	Kevin M	Normal	
949854	3/10/26 9:49 PM	New hire laptop request	Lisa P	Normal	
723851	3/10/26 9:48 PM	Request to install Adobe Acrobat Pro	Tom B	Normal	
193373	3/10/26 9:48 PM	Unable to connect to network drive after ...	Kevin M	Normal	
900476	3/10/26 9:47 PM	Monitor not displaying image on seconda...	Lisa P	Normal	
373650	3/10/26 9:46 PM	Locked out of Windows account after fail...	Tom B	Normal	

2a. Standard Tier 1 Resolution - Account Lockout

Ticket	Locked out of Windows account after failed login attempts
Submitted by	Tom Bradley
Help Topic	Password Reset / Account Lockout
Routed to	Tier 1 Support | SEV-3 | Assigned: Jake Morris

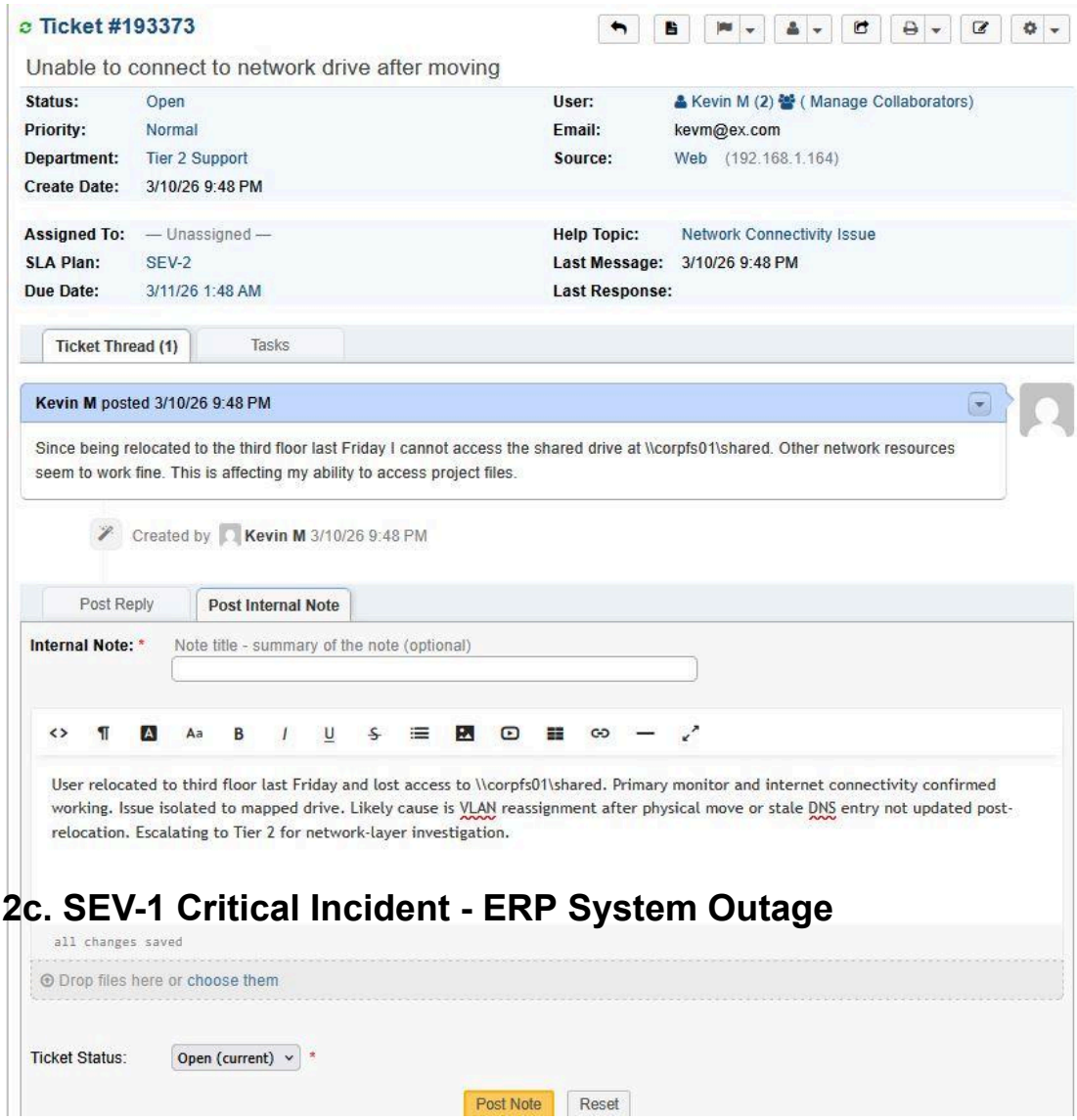
This ticket represents the most common Tier 1 category in a corporate environment. The account lockout was resolved at first contact without escalation. The agent replied directly to the user with resolution confirmation and closed the ticket.



2b. Tier 1 to Tier 2 Escalation - Network Drive Access

Ticket	Unable to connect to shared network drive after office move
Submitted by	Kevin Marsh
Help Topic	Network Connectivity Issue
Routed to	Tier 2 Support | SEV-2 | Assigned: John Webb

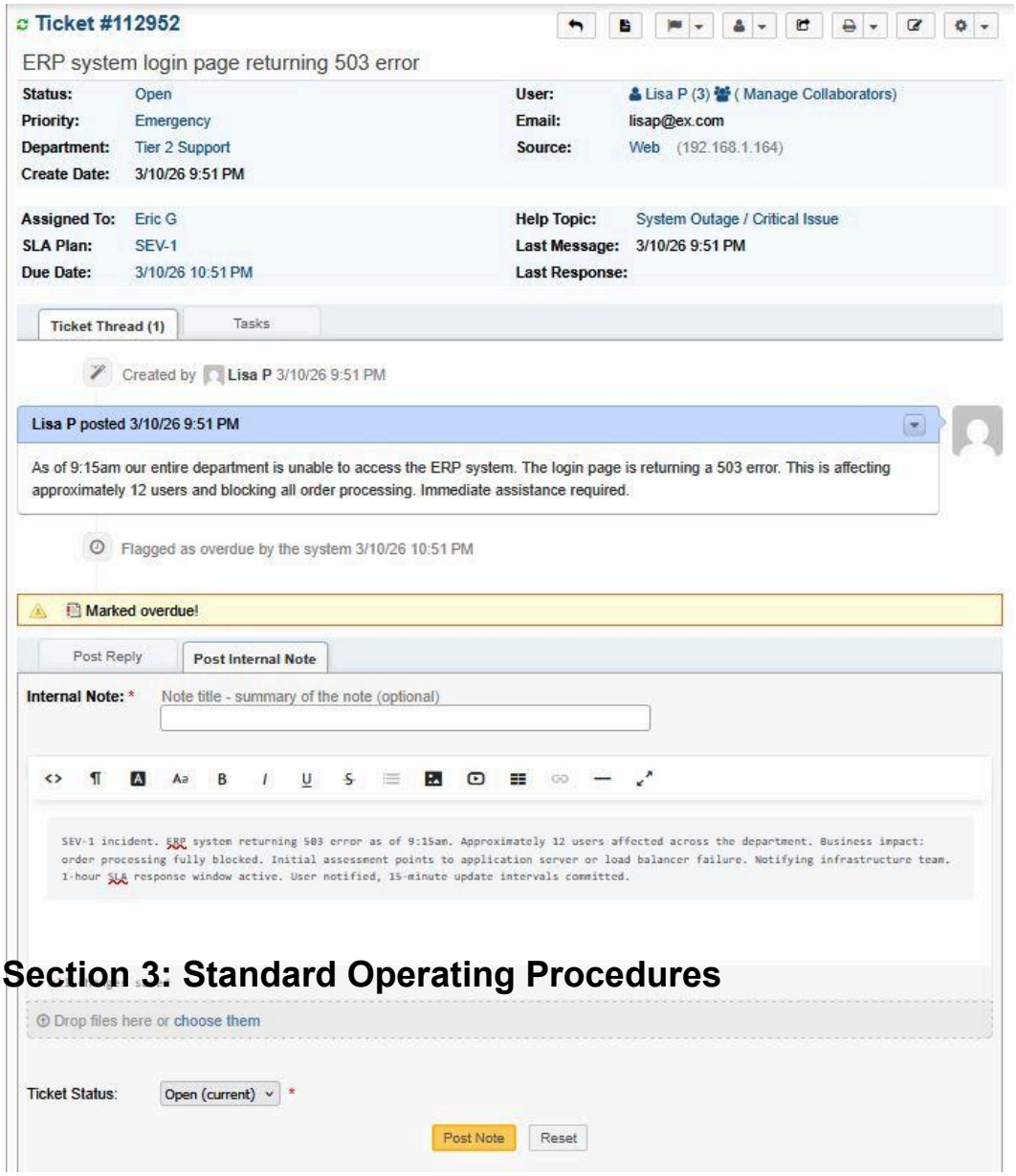
This ticket demonstrates the Tier 1 to Tier 2 escalation path. Initial triage identified a likely VLAN assignment or stale DNS issue resulting from the user's physical relocation. An internal note was posted documenting the working assessment before the ticket was transferred to Tier 2. The user received a reply acknowledging the escalation and setting expectations for follow-up. The receiving Tier 2 agent inherits the triage context from the note without needing to re-interview the user.



2c. SEV-1 Critical Incident - ERP System Outage

Ticket	ERP system login page returning 503 error for entire department
Submitted by	Lisa Park
Help Topic	System Outage / Critical Issue
Routed to	Tier 2 Support | SEV-1 | Assigned: John Webb

This ticket triggered SEV-1 priority routing automatically at submission based on the help topic configuration. The incident affected approximately 12 users and blocked all order processing. The internal note documented the time of detection, scope of impact, and the 1-hour SLA response window.



Section 3: Standard Operating Procedures

The following SOPs document the procedures used for each major ticket workflow category. These are written as working reference documents, not installation guides. Each SOP covers intake criteria, required actions, escalation conditions, and resolution standards.

SOP-01: Account Lockout and Password Reset

Scope	Active Directory account lockouts, password reset requests, login failures
SLA	SEV-3, 8-hour response during business hours
Assigned to	Tier 1 Support

1. Confirm the user's identity before making any account changes. Request employee ID or manager confirmation if identity cannot be verified through the ticket alone.
2. Unlock the account in Active Directory Users and Computers. Do not reset the password unless the user explicitly requests it or the account is in a broken state.
3. Confirm the unlock with the user by reply. If the user cannot log in after unlock, collect the exact error message and escalate to Tier 2.
4. Document the source IP or workstation name if available. Repeated lockouts from the same account may indicate a credential issue on a connected device and should be noted in the ticket.

Escalation criteria: Escalate to Tier 2 if the account continues to lock after reset, if multiple accounts are locking simultaneously, or if the pattern suggests a security incident rather than a user error.

SOP-02: Tier 1 to Tier 2 Escalation

Scope	Any ticket requiring investigation beyond Tier 1 resolution capability
SLA	Inherits from ticket; SEV-2 or above on escalation
Assigned to	Tier 1 initiates, Tier 2 receives

1. Before escalating, document all triage steps already taken in an internal note. The Tier 2 agent should not have to repeat work already performed.
2. Include in the internal note: the problem description in technical terms, what was already attempted, your working hypothesis for root cause, and any relevant system information gathered from the user.
3. Transfer the ticket department to Tier 2 Support and assign to the appropriate Tier 2 agent.
4. Post a reply to the user acknowledging the escalation and setting an expectation for follow-up. Do not leave the user without communication when a ticket changes hands.
5. The receiving Tier 2 agent should review the internal note and post an acknowledgment note confirming they have picked up the ticket.

Escalation criteria: Escalate when the issue requires network-layer access, server-side investigation, cannot be resolved within the Tier 1 SLA window, or when the scope expands to affect multiple users.

SOP-03: Critical Incident Response (SEV-1)

Scope	System outages, service failures affecting multiple users, business-blocking incidents
SLA	SEV-1, 1-hour response, 24/7 schedule
Assigned to	Tier 2 Support

1. Immediately confirm SEV-1 SLA is applied. The 1-hour clock starts at ticket creation time, not at assignment.
2. Post an internal note within 15 minutes of pickup documenting: time of detection, number of affected users, systems involved, business impact, and initial hypothesis.
3. Send an acknowledgment reply to the reporter within 15 minutes. Commit to update intervals. Do not go silent on a critical incident ticket.
4. Notify the Infrastructure team if the incident involves server or network infrastructure. Notify IT Administration if account or access systems are involved.
5. Post a resolution note when service is restored documenting: time of restoration, root cause if known, and any temporary mitigations applied.

6. Do not close the ticket at resolution. Leave it open for a post-incident review period of at least one business day.

SOP-04: Hardware Provisioning Request

Scope	New hire equipment, replacement hardware, peripheral provisioning
SLA	SEV-3, 8-hour intake response. Delivery timeline varies by request type.
Assigned to	Infrastructure

1. On receipt, post an internal note confirming hardware inventory availability. If inventory is insufficient, flag immediately and notify the requestor.
2. Collect all required information before beginning imaging: employee full name, job title, department, required software beyond standard load, VPN access requirement, and start date.
3. Image the device using the standard build for the employee's department. Document the device serial number, assigned hostname, and imaging date in the ticket.
4. Confirm with the requestor when the device is ready for pickup or delivery. Do not close the ticket until the requestor confirms receipt.

SOP-05: Access and Permissions Request

Scope	Shared folder access, application permissions, group membership changes
SLA	SEV-3, 8-hour response. Processing within 1 business day of approval confirmation.
Assigned to	IT Administration

1. No access change is made before manager approval is confirmed in writing. Verbal approval is not sufficient.
2. Reply to the requestor requesting manager confirmation via ticket reply or forwarded email. Document in an internal note that approval is pending.

3. Once approval is confirmed, post an internal note recording the approver name, date, and method of confirmation before making any changes.
4. Apply the minimum necessary permissions to fulfill the request. Do not grant broader access than what was explicitly requested and approved.
5. Confirm the change with the requestor and document the exact permissions granted in the ticket before closing.

Section 4: Findings and Conclusions

4a. Configuration Summary

Component	Detail	Status
osTicket deployment	nginx + PHP-FPM + MySQL 8.4 on bare-metal Ubuntu 25.04. Fully operational.	Complete
Department structure	4 departments configured with SLA assignments and agent provisioning.	Complete
SLA tiers	SEV-1 (1hr/24x7), SEV-2 (4hr/24x7), SEV-3 (8hr/business hours).	Complete
Help topic routing	8 topics mapped to departments and SLA tiers. Routing confirmed functional.	Complete
Ticket workflows	8 tickets processed across 5 workflow categories. Escalation paths confirmed.	Complete
SOP documentation	5 SOPs written covering all major workflow categories.	Complete

4b. Analyst Notes

The most operationally significant configuration decision in this deployment is the help topic routing structure. By mapping each topic to a specific department and SLA at creation time, the system eliminates the need for a human triage step on standard ticket categories. Tickets arrive in the correct queue with the correct SLA already applied. This is how mature help desk environments are expected to operate: routing is a configuration problem, not a daily judgment call.

The escalation workflow between Tier 1 and Tier 2 depends entirely on internal note discipline. The value of the internal note is not just documentation for the record; it is the primary communication channel between the agent handing off the ticket and the agent receiving it. A transfer without an internal note forces the receiving agent to re-triage from scratch, which defeats the purpose of tiered support. The SOPs reflect this by requiring a documented assessment before any escalation.

The SEV-1 workflow highlights a distinction between ticket priority and ticket urgency that is easy to conflate. Priority is a configuration value that determines SLA and routing. Urgency is an analyst's judgment about how actively an incident needs to be worked. A SEV-1 ticket that has been acknowledged and is being actively investigated is different from a SEV-1 ticket sitting unassigned. The SOP addresses this by requiring acknowledgment within 15 minutes and committing to update intervals, not just setting the priority field and moving on.

4c. What I Would Do Next

The most immediate gap in the current deployment is email intake. osTicket 1.18.3 requires the OAuth2 plugin for Gmail integration because Google deprecated basic authentication for IMAP and SMTP. Configuring the OAuth2 plugin requires a Google Cloud project with the Gmail API enabled and an OAuth2 credential set. Once configured, inbound emails to the help desk address create tickets automatically and agents can reply through osTicket with responses delivered from the Gmail address. This is the primary intake channel in most corporate environments and is the next planned addition to this deployment.

The second gap is canned responses. osTicket supports predefined reply templates that agents can insert when working tickets. For high-volume categories like password resets and software requests, canned responses reduce handle time and standardize the language agents use when communicating with end users. Building a library of canned responses for each SOP category would be a direct extension of the workflow documentation already in place.

Longer term, the right addition to this environment is Wazuh integration. Wazuh is already deployed separately in this homelab. Connecting the two systems would allow security-related tickets to be opened automatically from Wazuh alerts, creating a documented response trail for each detection event. That integration would bridge the gap between the SOC monitoring environment and the help desk operations environment, reflecting how security operations and IT support interact in a mature organization.